Slide 1
CS 170
Java Programming 1
"In-Class" Labs
Duration: 00:01:18
Advance mode: Auto

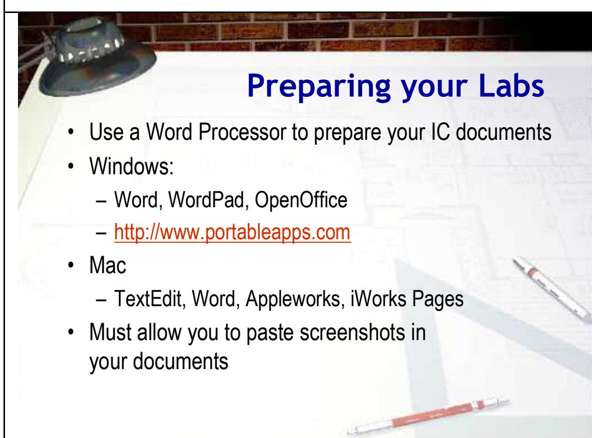
Notes:

Hi Folks. This is the CS 170, Java Programming 1 lecture on "In-Class" Labs

CS 170, as you know, is a lecture-lab class. Unlike in Biology or Chemistry, where you have a separate lab period, the CS 170 lab is integrated with the lecture. This has a couple of advantages, but the most important one is that you get some hands-on experience right when we're discussing a topic, instead of waiting until you try to do your homework. That way, if you have any problems, you have a chance to ask questions and correct your misconceptions earlier, instead of later.

For the in-person classes, (as you've already found out), I'll have you complete an "in-class" (IC) exercise document with screen-shots and answers to questions, which you'll turn in as a PDF file before you leave class. You'll do the same thing for the online portion of the class: in addition to your homework assignments, you'll need to answer the questions and do the exercises presented in the online lecture and submit them each week.

Since many of you are working at home, you'll need to assemble some tools to do that. This mini-lecture will walk you through the process of creating your weekly IC "in-class" lab exercise document.



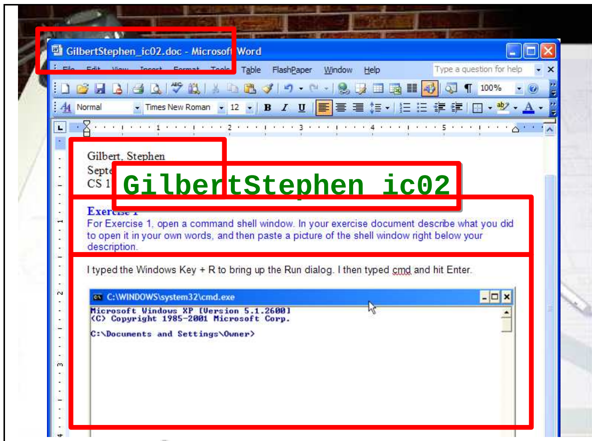
Slide 2
Preparing your Labs
Duration: 00:00:59
Advance mode: Auto

Notes:

Let's take a look at the tools you'll need to prepare and submit your weekly in-class lab documents. First, you can use any word processor you like on any platform to create the homework document. On Windows, for instance, you can use Microsoft Word if you have it installed. If you don't then you can use the WordPad program, which comes with all versions of Windows, or (and this will make your life so much easier), you can download OpenOffice. PortableApps.com has a portable version of OpenOffice for Windows that fits on a USB thumb drive. I really recommend this. If you have Linux, OpenOffice is probably already installed.

On the Mac, you can use Word, Appleworks, or iWork pages. The only requirement is that your word processor allow you to paste the screenshots of your work into your document.

Now, let's look at the requirements for the lab assignments.



Slide 3

IC Lab Requirements

Duration: 00:02:19
Advance mode: Auto

Notes:

1. Name your document using your last name followed by your first name. You must use the name that is on my class roster. Follow that with an underscore (not a hyphen), and the words ic01 in lower case. So, in my case, the first in-class exercise was GilbertStephen_ic01. The following lab exercises will be ic02, ic03, etc. Most of the weeks you'll submit two IC documents: one in class and one online. The online exercise page will give you the exact number for that week's online IC document. It's very important that you follow these naming conventions exactly, since your assignments are automatically routed by my grading software.
2. At the top of every homework assignment, you need to put the same four lines. The first line is your name. Put your last name first as I've done here. The second line should be the date that you started the assignment. The third line should identify the submission. "CS 170 In-Class Exercise 01", etc. Follow these three lines with a blank line.
3. As you complete each portion of the homework, make sure you type in the number that I've assigned for each problem, followed by a short description or your understanding about what you were supposed to do. You don't have to type the description verbatim; a short paraphrase is fine. One easy way to do this is to use the PDF lecture transcripts and just copy the text from the PDF. It will also help me if you use color or the font to set off the question from the answer, so that it's easy for me to read.
4. If the problem requires a verbal response, put your answers on the next line, using formatting (perhaps a bulleted list), so it's easy for me to see the difference between the question and your response. To show me that you've completed a particular task, I'll almost always ask you to show me a screenshot. Paste your screenshot below the problem or exercise, leaving a blank line between the text above the picture and another blank line following the picture.

Now let's tackle the problem of creating the screen-shots.

Slide 4

Making Screenshots

Duration: 00:02:21
Advance mode: Auto

Notes:

Making your screenshots is also easy, but many students don't know how to do it. Here's a short guide.

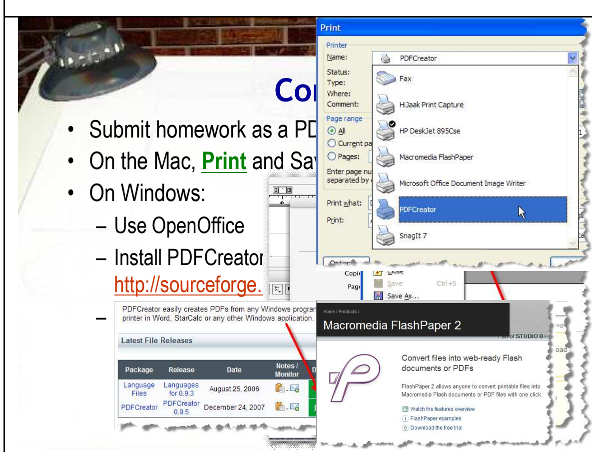
On the Mac use Command-Control-Shift-4,and then select an area with your mouse to take a screenshot of that area and save it to the clipboard. If you want to automatically select a window, press Command-Control-Shift-4 and then the space bar. The cursor will change to a camera and when you click a window you'll take a screenshot of a window and save it to the clipboard. Once you take the screenshot, transfer to your homework document and paste it at the appropriate location. If you want to take a whole lot of screenshots at once and then paste them into your homework document later, use the same keystrokes, but leave out the Control key; just use Command-Shift-4.

On Windows, select the window you want to display, hold down the Alt key and then press Print Screen. (The Print Screen key should be in the upper-right corner on your keyboard.) This copies the selected window to the clipboard where you can paste it into your word-processing document. There's no easy way to automatically save as a screenshot as a file on Windows.

Commercial screen capture software is fairly inexpensive and has a few more features, like the ability to annotate your captures. Like a lot of folks, I use SnagIt-Pro on Windows for my captures. I think that it's about $40. If you're interested, you can download a free trial version from Techsmith.

Recently, Techsmith introduced a new product called "Jing", that is kind of a cross between SnagIt "lite" and their screen-recording software "Camtasia". Unlike SnagIt and Camtasia, though, Jing works on both the PC and on the Mac. Jing isn't exactly a "product" though; Techsmith calls it a "pilot project". At the time I'm writing this, Jing is completely free, but that might not be the case when you read this.

If none of these solutions work for you, head on over to this Wikipedia page for everything you wanted to know about screenshots.



Notes:

While you can use any word processing format to create your lab documents (as well as some homework assignments), you must submit them in Portable Document Format or PDF. I won't accept Word or other documents.

If you're on a Mac, creating PDF files is really easy. You just print the file, and on the print dialog you click the Save as PDF button as shown here. Use the same filename as you used to create the document, but be sure to manually add the .pdf extension. It's not needed on the Mac, but once you turn it loose on the Internet it will help identify the type of data the file contains.

Slide 5 **Converting to PDF** Duration: 00:02:18 Advance mode: Auto	On Windows, you have several options since PDF creation isn't built into the operating system. • You can use OpenOffice, which directly exports to PDF using the menu selection or toolbar shown here. This is what I do, using the portable version of OpenOffice on my USB thumb drive, even if I've created the file in Microsoft Word. Just make sure, if you use Word 2007, that you save it as a .doc, not a .docx. • You can install a free PDF printer driver called PDFCreator. Get it from http://sourceforge.net/projects/pdfcreator. (Don't be fooled by the ads on the site for PDFCreator.com and for Adobe.) This allows you to create PDFs of anything you can print. I use it to create all of the PDFs for this class, and I've found that it works quite well. To use it, you just select PDFCreator as your printer when printing • You can use commercial PDF creation software such as Adobe Acrobat (which is quite expensive) or Adobe FlashPaper which is cheaper and can create Flash "printouts" as well. You'll turn in the in-person in-class exercises before you leave class; you'll submit the online in-class lab exercises electronically. Check the class Web page for the exact details about how to do that.

